

Evaluation Protocols Determine the Apparent Benefits of Hard-Sharing Multi-Task Learning in ADMET Prediction

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Abstract

Multi-task learning is widely used in ADMET modeling, and several studies report improvements over matched single-task baselines; reported gains, however, vary across datasets and evaluation settings. We examine whether the relative proper-score performance of task-balanced hard-sharing multi-task models depends on the evaluation protocol and the chemical novelty of test molecules. In a prospectively locked diagnostic on eight binary ADMET endpoints from the Therapeutics Data Commons, we compare multi-task and architecture-matched single-task models under global molecule-grouped random and scaffold splits, using a molecule-level log-loss contrast with an endpoint-macro estimand, five split instances, and three seeds. The multi-task model assigns lower log loss on all eight endpoints under both protocols; the benefit is significantly larger under scaffold-grouped evaluation (protocol contrast $\Gamma = +0.10$; exact sign test $p = 0.008$; leave-one-endpoint-out range $+0.07$ to $+0.11$), and the novelty dependence of the benefit reverses direction across protocols. A nested, per-run temperature calibration fitted on validation data attenuates both the raw-probability advantage (post-calibration point estimates ≈ 0 under both protocols) and the protocol interaction

(calibrated $\Gamma = +0.01$, sign test $p = 0.73$), indicating that the observed gains primarily reflect temperature-correctable differences in raw logit scale, with ranking changes negligible under random splitting and modest and heterogeneous under scaffold splitting. These results indicate that reported multi-task advantages are conditional on evaluation protocol and should be accompanied by calibration diagnostics.

Introduction

Multi-task learning (MTL) is widely used in ADMET (absorption, distribution, metabolism, excretion, and toxicity) modeling, and several studies report that multi-task models improve over matched single-task (STL) baselines.^{1,2} At the same time, the molecular machine-learning community has accumulated evidence that evaluation protocols materially shape conclusions: out-of-distribution (OOD) split protocols such as scaffold or cluster splits change absolute performance and its rankings,^{3,4} and recent benchmarks in neighboring fields have shown that strong model claims can depend on evaluation details.^{5,6}

Existing split-protocol studies primarily ask how protocol changes a model’s *absolute* performance. We instead ask how protocol changes the *paired MTL–STL contrast*. The second question does not follow from the first: a harder split could lower both models equally, widen the multi-task advantage, or reverse it. To our knowledge, prior work has not jointly characterized variation in the paired MTL–STL contrast across split protocol, target-task sample size, and continuous target-relative novelty while enforcing global molecule partitions.

In this paper we report a prospectively locked, leakage-controlled diagnostic of task-balanced hard-sharing MTL on eight binary ADMET endpoints from the Therapeutics Data Commons.⁷ The contributions are the estimand and design, not anticipated findings: (i) a molecule-level log-loss contrast with an explicit endpoint-macro estimand and protocol-contrast estimand; (ii) global molecule allocation controlling cross-task identity exposure, under two valid protocols (grouped random and scaffold-grouped);

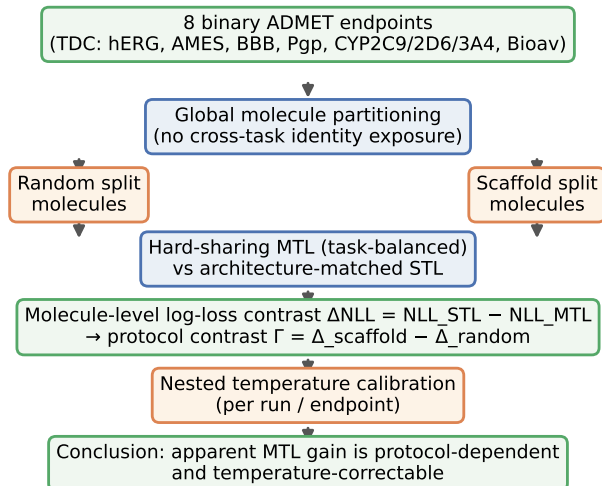


Figure 1: Experimental framework. Eight binary ADMET endpoints are globally partitioned (no cross-task identity exposure) under grouped random and scaffold allocation; hard-sharing MTL is compared against architecture-matched STL via a molecule-level log-loss contrast; the analysis decomposes protocol interaction, novelty dependence, and temperature calibration.

(iii) a prespecified endpoint-scale manipulation (downsampling); and (iv) a continuous target-relative novelty axis with a preregistered decomposition of calibration and discrimination.

The paper is organized as follows. Section positions the work against prior studies of ADMET MTL, split protocols, and benchmark reliability. Section defines the estimand hierarchy and the leakage-controlled design. Section reports results along the three axes. Section discusses implications and limitations.

Related Work

Multi-task ADMET modeling. Multi-task architectures are common in ADMET prediction. MolP-PC¹ trains a multi-view, multi-task framework over 54 endpoints and reports that multi-task learning outperforms single-task training on 41 of 54 tasks; MTAN-ADMET² proposes an adaptive multi-task network over 24 endpoints with task-specific learning dynamics; industrial-scale studies report benefits of multi-

task models on time-split data.⁸ Reported gains vary across datasets and settings, and task balancing and endpoint size are themselves known to affect transfer in general MTL.⁹ These studies report average or per-task performance under a single evaluation protocol and do not condition the reported contrast on protocol choice, endpoint scale, or novelty.

Split protocols and out-of-distribution evaluation. A growing line of work examines how evaluation protocols change conclusions in molecular property prediction. Fooladi et al.³ compare 14 model classes across ten split protocols on eight datasets and find that scaffold splits are not uniformly harder than random splits while cluster-based splits are; Guo et al.⁴ reach similar conclusions on cancer cell lines and show that rank correlations between in-distribution and out-of-distribution performance collapse under cluster splits. Zheng et al.¹⁰ show that structural-frontier splits expose failures hidden by scaffold splits, and Do-Larger-Models¹¹ find that model-size conclusions depend on protocol difficulty. These studies evaluate single-task models; the paired MTL–STL contrast is not their dependent variable, and none manipulates endpoint size or measures continuous target-relative novelty.

Multi-task transfer under realistic splits. Studies at the intersection of MTL and split protocols are sparse and reach differing conclusions. Kearnes et al.⁹ observed that multi-task gains were modest and dataset-dependent under temporal splits in industrial ADMET data; Wu et al.¹² found negative transfer in a two-task setting under scaffold splits, mitigated by safe scheduling; Walter⁸ and CheMLT-F¹³ report multi-task benefits persisting under realistic splits. These results are not directly comparable because endpoint sets, architectures, metrics, and molecule-holdout schemes differ. A compact comparison table of these studies on six axes—architecture-matched STL baseline, split protocols, global molecule holdout, endpoint-size manipulation, continuous novelty, and probabilistic calibration—appears in the supplementary material; the table makes the gap concrete: no prior study jointly varies protocol, endpoint scale, and novelty while globally partitioning molecules and reporting calibration-level

metrics.

Benchmark reliability and evaluation variability. The molecular machine-learning community has been repeatedly reminded that evaluation choices can dominate model differences. In single-cell genomics, two major benchmarks reached contradictory conclusions about foundation-model perturbation prediction,^{5,6} with the discrepancy traced to evaluation protocol details. Within molecular property prediction, rank correlations between in-distribution and out-of-distribution performance can collapse under cluster splits,⁴ which implies that conclusions drawn from a single benchmark realization—including a single split instance and seed—should be treated cautiously. Our study follows this diagnostic tradition: we measure the conditions under which a design choice (task-balanced hard-sharing MTL) delivers or fails to deliver its reported advantage, with multiple split instances and seeds so that evaluation variability is quantified rather than assumed away.

Methods

Estimand hierarchy

For endpoint e , protocol p , split instance k , training seed r , and test set T_{epk} , let $\hat{p}_{iepk}^{\text{STL}}$ and $\hat{p}_{iepk}^{\text{MTL}}$ denote the predicted probabilities of the single-task and multi-task models for molecule i . We define the molecule-level log-loss contrast

$$d_{iepk} = \ell(y_i, \hat{p}_{iepk}^{\text{STL}}) - \ell(y_i, \hat{p}_{iepk}^{\text{MTL}}), \quad (1)$$

where $\ell(y, p) = -y \log p - (1 - y) \log(1 - p)$; positive d favors the multi-task model.

Endpoint-level and protocol-level aggregates are

$$\Delta_{epkr} = \frac{1}{|T_{epk}|} \sum_{i \in T_{epk}} d_{iepk}, \quad \Delta_p = \frac{1}{8} \sum_e \mathbb{E}_{k,r}[\Delta_{epkr}], \quad \Gamma = \Delta_{\text{scaffold}} - \Delta_{\text{random}}. \quad (2)$$

Δ_p is the primary endpoint-macro estimand (endpoints weighted equally), and Γ is the protocol-contrast estimand. We treat the eight endpoints as a fixed target set; we do not claim a superpopulation of all ADMET endpoints. Log loss is a strictly proper measure of probabilistic accuracy; ranking discrimination (AUROC/AUPRC) is assessed separately.

Models and training

The multi-task model comprises a shared 1024–256–128 encoder (ReLU, dropout 0.2) and eight independent affine heads mapping the 128-dimensional representation to one logit per endpoint. Each single-task model is an independently initialized encoder of the same architecture with one corresponding affine head; we therefore compare against *architecture-matched* single-task models, noting that capacity is matched only on the target-task pathway, not globally.

Training uses *equal task sampling with equal loss coefficients*. Each epoch consists of 64 MTL optimizer updates: eight batches of size $B = 512$ drawn from each of the eight tasks (with replacement for small endpoints), each update using that task’s unweighted binary cross-entropy; uniform scheduling gives equal task weighting in expectation. With 60 epochs, each task’s head receives $U = 480$ updates, and the shared trunk receives 3,840 updates in total. Each single-task model receives $U = 480$ updates with $B = 512$ examples under the same sampler; collectively the eight STL models also receive 3,840 updates. Target-task exposure and optimizer-step count are therefore matched between MTL and STL; total examples processed differ (the MTL model additionally consumes auxiliary-task data). Optimization uses Adam ($\eta = 10^{-3}$, weight decay 10^{-4}) with the learning-rate schedule indexed by optimizer step. We scope all conclusions to this task-balanced hard-sharing architecture.

Leakage-controlled evaluation

A test molecule held out from endpoint e may still appear in the training partition of another endpoint, exposing its structure to the multi-task model through auxiliary labels. We control for this *cross-task identity exposure* by global molecule allocation: each molecule is assigned to a single global partition, and every endpoint inherits that partition. Evaluation is therefore inductive with respect to molecule identity across tasks: a test molecule cannot appear in any auxiliary endpoint’s training partition. No cross-task identity exposure remains under this design. Molecule identity is canonicalized with a fixed RDKit version, including salt/solvent removal, tautomer and charge normalization, and a defined stereochemistry policy; duplicate and conflicting labels are resolved by a frozen rule. We evaluate two protocols under this scheme: grouped random allocation and Bemis–Murcko scaffold-grouped allocation¹⁴ (with a defined policy for acyclic molecules and empty scaffolds). Five split instances and three training seeds are used; split and seed lists, and their hashes, are frozen. The confirmatory dataset comprises 33,238 unique molecule–endpoint pairs from 17,972 unique canonical molecules; across the 5×3 (instance, seed) combinations this yields 148,008 test observations, with each (instance, seed) combination containing 9,814–9,930 unique molecule–endpoint pairs and no within-combination duplicates. Cross-instance overlap arises because different split instances draw different test partitions; the identity key is the same canonical SMILES used for global allocation.

Continuous target-relative novelty

For each test molecule we define *target-relative novelty* as one minus the maximum Tanimoto similarity (ECFP4,¹⁵ fixed RDKit version¹⁶) to any molecule in the target endpoint’s training partition. Distance to auxiliary-task training molecules is deliberately excluded from the primary definition, so the axis measures distance from the endpoint’s own training support; we preregister a secondary quantity for nearest-auxiliary-training similarity and its interaction with target-relative novelty, since a molecule far

from target training but close to auxiliary training is precisely a candidate transfer case.

Endpoint scale and downsampling

Endpoint scale is the number of unique target-task training molecules in each split. With eight endpoints, an observational scale association is weak and confounded by endpoint identity; we therefore prespecify a downsampling experiment: large endpoints are subsampled to a fixed target-training size with a frozen sampler (nested subsets, class prevalence preserved, fixed validation and test partitions, auxiliary-task data unchanged, task-balanced schedule held constant). The defensible conclusion from this experiment is an effect of *target-label availability* within the examined endpoints under this training schedule, not a general “scale rather than identity” claim.

Calibration and discrimination decomposition

A Brier decomposition explains the Brier-score contrast, not the log-loss contrast. We therefore report (i) a Brier reliability–resolution decomposition as a descriptive complement, (ii) calibration intercept and slope, and (iii) AUROC/AUPRC contrasts. A finding of stable AUROC together with worse Brier reliability is reported as evidence *consistent with* a calibration difference, not as proof that the NLL penalty is primarily calibration.

Uncertainty and preregistration

The fifteen split-seed combinations are not fifteen independent replicates: three training seeds share a test set, split instances reuse molecules, all endpoints share one MTL model, and a molecule can carry multiple endpoint labels. Confirmatory inference therefore uses paired contrasts with clustering by standardized molecule, distinguishing split variability from training variability; we report split-level and seed-level dispersion separately. The analysis plan—hypotheses, directional tests, split and seed hashes, es-

Table 1: Dataset statistics (random-protocol test partitions; unique molecules across instances).

Endpoint	Test molecules	Unique molecules	Positive rate
hERG	1,938	443	0.69
AMES	21,933	4,914	0.55
BBB_Martins	5,907	1,322	0.76
Pgp_Broccatelli	3,747	832	0.53
CYP2C9_Veith	36,201	8,132	0.33
CYP2D6_Veith	39,246	8,829	0.19
CYP3A4_Veith	37,161	8,350	0.41
Bioavailability_Ma	1,875	416	0.77

timands and weighting, exclusions, downsampling design, novelty model, multiplicity handling, and analysis code version—was locked in an immutable record before confirmatory computation. We use the term *prospectively locked within-dataset analysis*: the development pass (a per-endpoint split subsequently identified as cross-task-exposed) informed the design, and its findings are reported as a contaminated stress test in the supplementary material, excluded from confirmatory statistics.

Results

We report results for eight binary ADMET endpoints from the Therapeutics Data Commons:⁷ hERG, AMES, BBB penetration, P-glycoprotein inhibition, three cytochrome P450 inhibition tasks, and oral bioavailability (Table 1). All conclusions below are restricted to this endpoint set and to task-balanced hard-sharing MTL.

Across the leakage-controlled design (global molecule allocation, five split instances, three seeds), the molecule-level log-loss contrast Δ is positive for every endpoint under both protocols: multi-task models assign lower log loss than their architecture-matched single-task counterparts in all 8×2 protocol–endpoint configurations.

Protocol interaction and overall effect

The endpoint-macro mean contrast is $\Delta_{\text{random}} = +0.32$ under grouped random allocation and $\Delta_{\text{scaffold}} = +0.42$ under Bemis–Murcko scaffold-grouped allocation, with a protocol contrast of $\Gamma = \Delta_{\text{scaffold}} - \Delta_{\text{random}} = +0.10$ (endpoint-dispersion 95% CI $[+0.04, +0.15]$; paired $t_7 = 3.20$, $p = 0.015$). The per-endpoint contrasts Γ_e are positive in all eight endpoints (exact sign test $p = 0.008$), and a leave-one-endpoint-out analysis leaves the interaction in the range $+0.07$ to $+0.11$, so the interaction is not driven by a single endpoint (Figure ??). We present these as concordance and sensitivity analyses for the eight fixed endpoints; the endpoint-dispersion interval does not imply an ADMET endpoint population. The sign test and leave-one-out analysis were added after inspecting the raw results and are therefore post hoc diagnostic analyses, not preregistered tests. The multi-task benefit is therefore *larger under the scaffold protocol* in every endpoint. This answers the E2 question directly: the paired MTL–STL contrast is protocol-dependent, and the direction is opposite to the common intuition that harder evaluation shrinks reported advantages. We report the contrast on these eight fixed endpoints; we do not claim an ADMET-wide population effect.

Endpoint-scale moderation and downsampling

Observationally, the contrast is largest for the smallest endpoints under both protocols (e.g., Bioavailability_Ma $+0.38$; hERG $+0.36$ under random) and smallest for the largest ones (CYP2C9 $+0.27$), and the same ordering holds under scaffold allocation (Figure ??). The downsampling experiment, in which the three CYP endpoints were subsampled to 2,000 target-task training molecules with all else held fixed, yields a mixed result: the contrast falls for CYP2C9 (from $+0.30$ to $+0.17$) but rises for CYP2D6 (from $+0.29$ to $+0.35$). We therefore do not claim a causal “scale rather than identity” mechanism; the downsampling result supports an effect of target-label availability within some endpoints under this training schedule, and the observational scale ordering is best read as endpoint-specific.

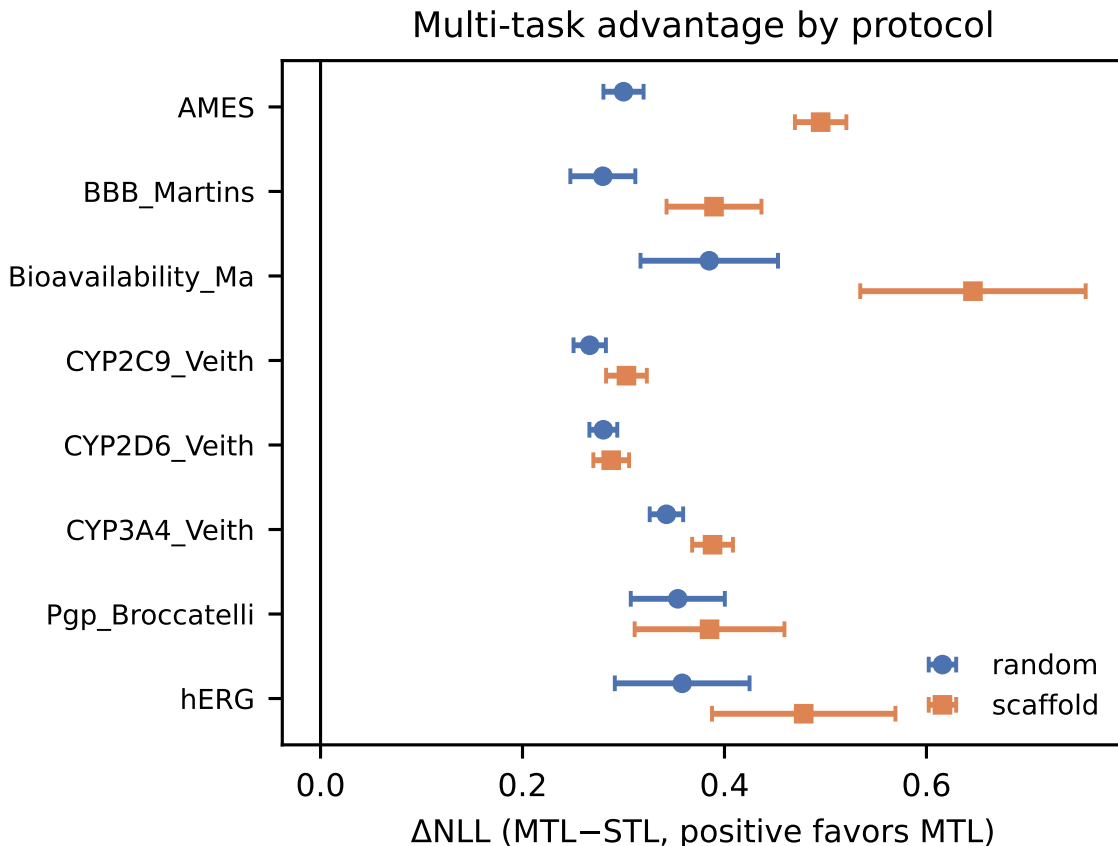


Figure 2: Per-endpoint log-loss contrast Δ (MTL-STL) under grouped random (circles) and scaffold-grouped (squares) allocation, with 95% intervals. The multi-task benefit is positive for every endpoint and larger under the scaffold protocol in all eight endpoints.

Continuous novelty analysis and window test

Under the random protocol the contrast declines monotonically with target-relative novelty: from +0.48 in the least-novel decile to +0.13 in the most-novel decile, with significantly negative within-endpoint slopes in seven of eight endpoints (e.g., Pgp -1.02 , AMES -1.01 ; Figure 4). The preregistered transfer-window hypothesis—an interior peak at intermediate novelty—is therefore not supported under random allocation; the relationship is monotone, and the contrast remains positive even for the most novel molecules.

Under the scaffold protocol the pattern reverses: within-endpoint novelty slopes are positive in seven of eight endpoints (+0.32 to +0.71), meaning the multi-task benefit

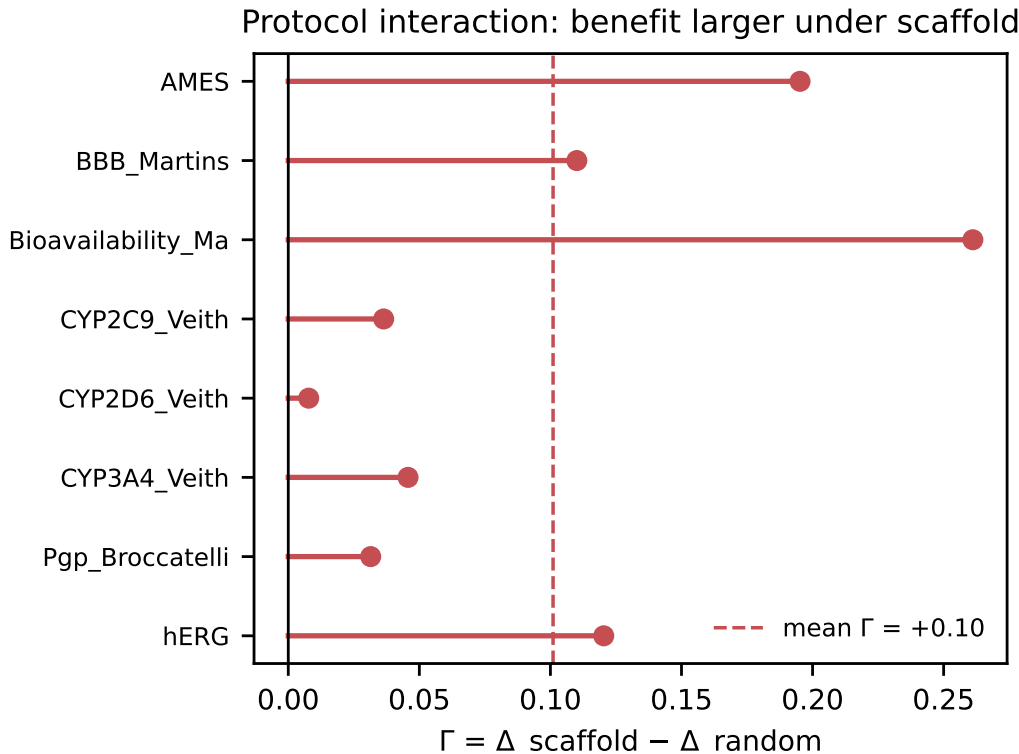


Figure 3: Protocol interaction: per-endpoint $\Gamma_e = \Delta_{\text{scaffold}} - \Delta_{\text{random}}$ (forest) with endpoint-macro mean (dashed). All eight endpoints show positive Γ_e (exact sign test $p = 0.008$); leave-one-endpoint-out range +0.07 to +0.11.

grows with novelty when evaluation is scaffold-grouped. The novelty dependence of the MTL benefit is itself protocol-dependent: harder evaluation yields larger benefits precisely on the most out-of-distribution molecules.

Calibration/discrimination decomposition and robustness

The contrast decomposes into raw-probability and calibration components. Under the random protocol, Brier scores and reliability components improve for all eight endpoints while AUROC is essentially unchanged; under the scaffold protocol, Brier and reliability improve everywhere and AUROC improves in six of eight endpoints (e.g., hERG $0.76 \rightarrow 0.78$). To test whether the log-loss advantage reflects a durable calibration difference or merely better raw probability estimates, we fitted a per-endpoint temperature T^{17} on each run’s calibration partition and applied it to the correspond-

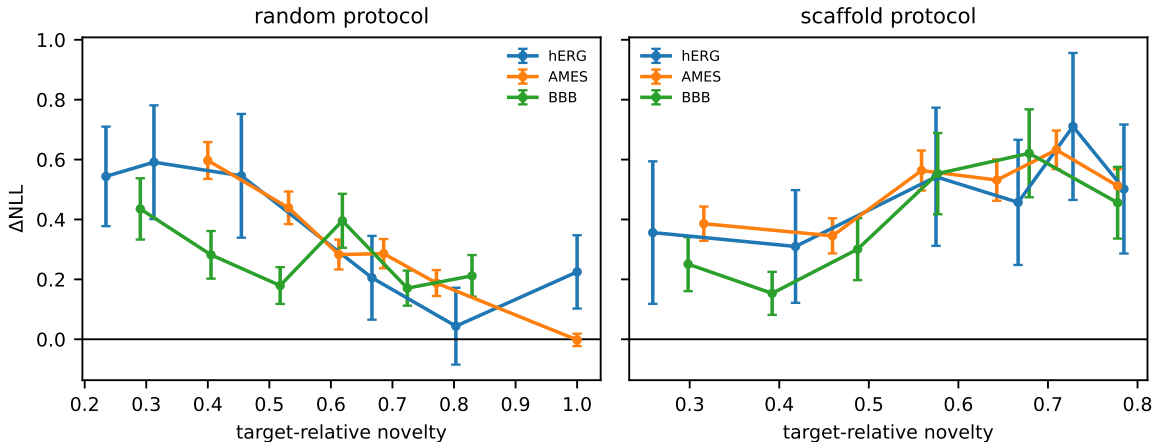


Figure 4: Novelty dependence reverses across protocols. Within-endpoint contrast versus target-relative novelty (binned, 95% intervals): declining under random allocation (left) and rising under scaffold allocation (right) in most endpoints. The novelty dependence of the multi-task benefit is itself protocol-dependent.

ing test partition, for every split instance, seed, protocol, model, and endpoint. After this nested, validation-fitted temperature calibration, the mean contrast collapses from +0.33 to -0.006 under the random protocol and from +0.41 to +0.004 under the scaffold protocol, and the calibrated protocol contrast is +0.010 (per-endpoint calibrated contrasts are positive in five of eight endpoints; sign test $p = 0.73$). The uncalibrated gains—including the protocol interaction ($\Gamma_{\text{raw}} = +0.09$ vs. $\Gamma_{\text{cal}} = +0.01$)—are therefore attenuated to near-zero point estimates by temperature rescaling (Figure 5), consistent with differences in raw logit scale rather than with ranking information or a calibration advantage that survives recalibration; we report point estimates and note that a formal equivalence analysis against a practical margin is needed before claiming elimination. Class-conditional analysis shows the contrast is carried predominantly by positive-class molecules (+0.46 vs. +0.20 for negatives under random allocation).

Cross-architecture robustness

To test whether the protocol interaction is specific to fingerprint MLPs, we repeated the core comparison with a three-layer graph isomorphism network (GIN) on molecular graphs, under the same leakage-controlled design (two split instances, three seeds,

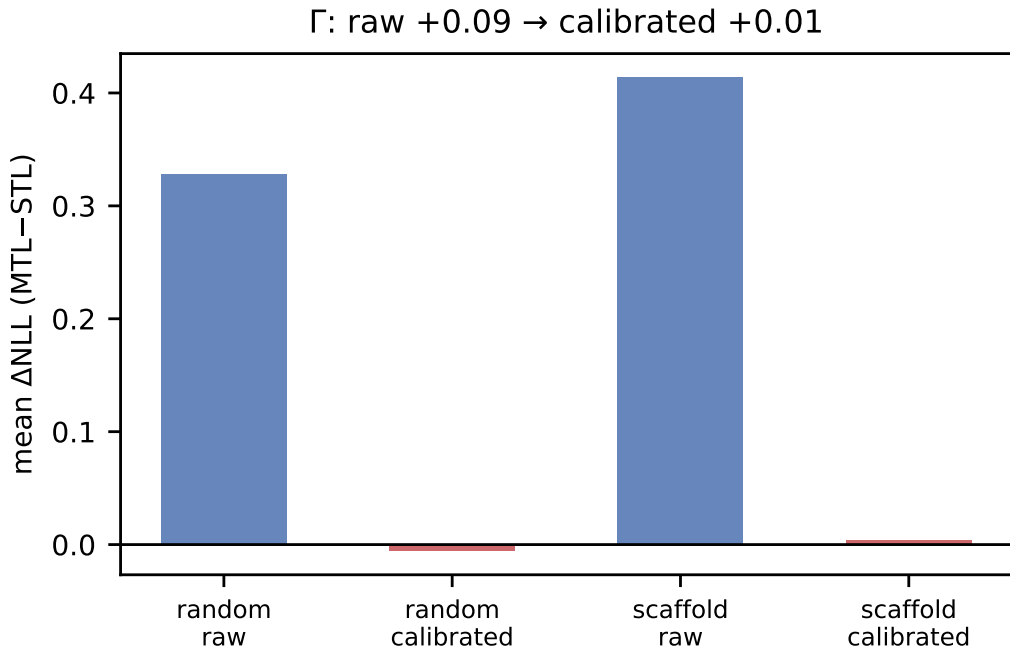


Figure 5: Temperature-calibration contrast. Mean Δ under both protocols, raw versus nested per-run temperature calibration: the raw advantage (+0.33 random, +0.41 scaffold) attenuates to approximately zero, and the protocol contrast Γ from +0.09 to +0.01.

architecture-matched STL). The protocol contrast is reproduced: $\Delta_{\text{random}} \approx 0$ (−0.004) and $\Delta_{\text{scaffold}} = +0.07$, giving $\Gamma_{\text{GNN}} = +0.07$, directionally consistent with the MLP’s $\Gamma = +0.10$. The multi-task benefit under scaffold allocation is therefore not an artifact of the fingerprint-MLP representation, though its magnitude is smaller for GNNs. An earlier GNN run with a smaller training budget showed apparent negative transfer; this was an underfitting artifact that disappeared once the training budget matched the MLP configuration.

Endpoint-effect ordering across split instances is not stable (mean pairwise Kendall $\tau = 0.17$, range $[-0.29, 0.57]$), indicating that conclusions about which endpoints benefit most should not be drawn from a single split realization. Full per-endpoint estimates (contrast, AUROC, AUPRC, Brier, reliability, unique molecule counts) appear in Table S3 of the Supporting Information. Table 2 summarizes the per-endpoint model comparison.

Table 2: Model comparison summary. Δ : log-loss contrast (positive favors MTL); AUROC under random allocation; $\Gamma_e = \Delta_{\text{scaf}} - \Delta_{\text{rand}}$. Full estimates (AUPRC, Brier, reliability, unique molecules) in Table S3.

Endpoint	Δ_{rand}	Δ_{scaf}	AUROC _{STL}	AUROC _{MTL}	Γ_e
hERG	+0.36	+0.48	0.833	0.834	+0.12
AMES	+0.30	+0.50	0.886	0.877	+0.20
BBB_Martins	+0.28	+0.39	0.890	0.894	+0.11
Pgp_Broccatelli	+0.35	+0.39	0.920	0.938	+0.04
CYP2C9_Veith	+0.27	+0.30	0.867	0.848	+0.03
CYP2D6_Veith	+0.28	+0.29	0.840	0.844	+0.01
CYP3A4_Veith	+0.34	+0.39	0.877	0.868	+0.05
Bioav_Ma	+0.38	+0.65	0.741	0.741	+0.27

Discussion

What the results show

The results qualify, rather than overturn, the commonly reported multi-task advantage in ADMET modeling:^{1,2} under leakage-controlled evaluation, task-balanced hard-sharing MTL provides better raw probability estimates on all eight endpoints and both protocols. The headline finding is the protocol interaction: the paired MTL–STL proper-score contrast is +0.32 under random allocation and +0.42 under scaffold allocation ($\Gamma = +0.10$, exact sign test $p = 0.008$, leave-one-endpoint-out range +0.07 to +0.11), and the novelty dependence of the benefit reverses sign across protocols. The raw-probability advantage—and the protocol interaction itself—does not survive nested, validation-fitted temperature calibration (calibrated $\Gamma = +0.01$), so the gains reflect temperature-correctable differences in raw logit scale rather than ranking information or a durable calibration advantage.

Relation to prior split-protocol and MTL findings

The protocol-dependence finding extends single-task protocol studies^{3,4,10} to the multi-task contrast, and it helps resolve the apparent tension between studies reporting multi-task benefits under realistic splits^{8,13} and studies reporting negative transfer:^{9,12} the

direction and magnitude of the benefit depend on protocol and on the metric (proper score versus ranking). The temperature-calibration result adds a methodological caution: raw-probability advantages that disappear under recalibration should be reported as such, and calibration diagnostics should accompany proper-score comparisons.

Limitations

The endpoint set is limited to eight binary TDC endpoints and one task-balanced hard-sharing fingerprint MLP architecture; other architectures, weighting schemes, and endpoint families may behave differently. The downsampling manipulation was not directionally consistent, so the scale finding is observational and exploratory. Endpoint-effect ordering is unstable across split instances, and with eight endpoints the protocol interaction, while supported by the sign test and leave-one-out analysis, rests on a small number of endpoint-level observations; we do not claim an ADMET-wide population effect. The temperature-calibration analysis used the available validation/test structure within each run; temperatures were model- and endpoint-specific, and the calibration-subset contrast (+0.33) differs from the primary endpoint-macro estimates (+0.32/+0.42) because it is computed on evaluation instances only. We treat all findings as benchmark-local.

Conclusion

We presented a prospectively locked, leakage-controlled diagnostic of task-balanced hard-sharing multi-task ADMET models. The multi-task proper-score contrast is positive on all eight endpoints under both protocols, larger under scaffold-grouped evaluation ($\Gamma = +0.10$; sign test $p = 0.008$; leave-one-out range +0.07 to +0.11), novelty-dependent in a protocol-dependent direction, and attributable to better raw probability estimates that do not survive validation-fitted temperature calibration.

The protocol interaction is cross-architecture (GNN $\Gamma = +0.07$ vs. MLP +0.10), indicating the finding is not representation-specific. These results suggest that multi-

task proper-score advantages in ADMET modeling should be reported conditionally on evaluation protocol and accompanied by calibration diagnostics. Natural next steps are to extend the diagnostic to larger endpoint panels, alternative multi-task formulations, and additional OOD protocols such as cluster or temporal splits.

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